

# NOTES 19: CLT-BASED INFERENCE FOR PROPORTIONS

Stat 120 | Spring 2026

Prof Amanda Luby

| Name                      | Type of Variable(s)                                      | Statistic               | Parameter       |
|---------------------------|----------------------------------------------------------|-------------------------|-----------------|
| Mean                      | Quantitative                                             | $\bar{x}$               | $\mu$           |
| Proportion                | Categorical                                              | $\hat{p}$               | $p$             |
| Standard Deviation        | Quantitative                                             | $s$                     | $\sigma$        |
| Difference in Proportions | 2 Categorical<br>(1 Response, 1 Explanatory)             | $\hat{p}_1 - \hat{p}_2$ | $p_1 - p_2$     |
| Difference in Means       | 1 Quantitative (Response)<br>1 Categorical (Explanatory) | $\bar{x}_1 - \bar{x}_2$ | $\mu_1 - \mu_2$ |
| Correlation               | 2 Quantitative                                           | $r$                     | $\rho$          |
| Slope                     | 2 Quantitative                                           | $b_1$                   | $\beta_1$       |

**CLT:** The Central Limit Theorem (CLT) tells us that if the sample size is big enough and the sample is random,

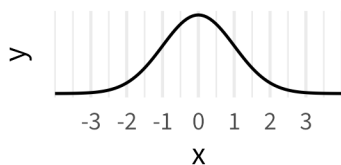
$$\bar{X} \sim N(\text{_____, ____})$$

$$\hat{p} \sim N(\text{_____, ____})$$

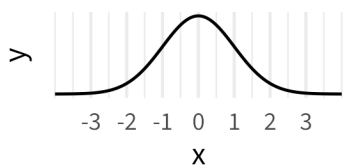
The general form for a **confidence interval** is:

**Example:** Finding  $z^*$

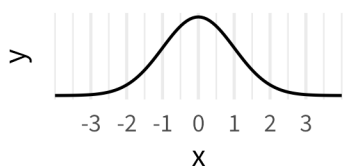
- 95% confidence interval



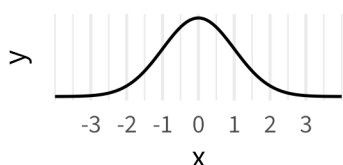
- 68% confidence interval



- 99% confidence interval



- 90% confidence interval



## 0.1 How big is big enough?

**Example 1:**  $\hat{p} = .5$

**Example 1:**  $\hat{p} = .05$

### Rule of Thumb for Proportions:

- Expected count in each category (Yes/No) should be  $> \rule{1cm}{0.4pt}$
- $np > \rule{1cm}{0.4pt}$  and  $n(1 - p) > \rule{1cm}{0.4pt}$

## 0.2 How do we find the SE?

### Standard Error for Proportions:

**Idea:** As  $n$  gets bigger, the SE gets  $\rule{1cm}{0.4pt}$ .

If  $\hat{p}$  is close to .5, the SE is  $\rule{1cm}{0.4pt}$  than if  $\hat{p}$  is close to 0 or 1

### Example: ESP Example Again

n = 14

```
p_hat = 3/14
p_0 = .2
n = 14
SE_p = sqrt((p_0*(1-p_0))/n)
SE_p
```

[1] 0.1069

```
z_score = (p_hat - .2)/SE_p
z_score
```

[1] 0.1336

```
p_val = pnorm(z_score, lower.tail = FALSE)
p_val
```

[1] 0.4468

n=1400:

n=14000:

**Big Picture** Picture

## 1 In-class Group Questions

1. APM Research Lab ran a [survey of likely Minnesota voters](#) between Sept 16-18 of 2024. They reported 48.4% in support of the Harris/Walz ticket, and 43.3% in support of the Trump/Vance ticket.

Here's an excerpt from their methodology report:

*The margin for error, according to standards customarily used by statisticians, is no more than  $\pm 3.5$  percentage points. This means that there is a 95% probability that the "true" figure would fall within that range if all voters were surveyed.*

- (a) Show how the margin of error was computed
  - (b) Is it OK to use the normal distribution as a model for the sampling distribution of a proportion in this case?
- 
2. In a random sample of 100 moviegoers in January 2013, 64 of them said they are more likely to wait and watch a new movie in the comfort of their own home.
    - (a) Is it ok to use the normal distribution as a model for the sampling distribution of the proportion of moviegoers preferring to wait and watch at home?
    - (b) Suppose I want to know if more than 50% of moviegoers in the population prefer to wait and watch at home. Write down the corresponding null and alternative hypotheses.
    - (c) For the hypothesis test in part (b), find the standardized test statistic (i.e., the Z-score).

(d) Calculate the p-value and state a conclusion in context

(e) The table below gives some of the percentiles of the  $N(0,1)$  distribution. If you were given this table (for example on an exam, where you do not have access to R), what could you say about the value of the p-value? (*Hint: draw the picture*)

| percentage | percentile (qnorm(percentage)) |
|------------|--------------------------------|
| 90%        | 1.3                            |
| 95%        | 1.6                            |
| 97.5%      | 2.0                            |
| 99%        | 2.3                            |
| 99.5%      | 2.6                            |

(f) Compute and interpret a 90% confidence interval for the population proportion of moviegoers who believe they are more likely to watch a new movie from home. (Use the information from the previous table to find the critical value needed.)

(g) How big of a sample size would you need if you wanted the margin of error for a 90% confidence interval to be no more than 2%? To answer this question, assume that the true proportion is  $p = 0.5$ , which would result in the biggest margin of error for a given sample size.